Exercise Solution

problem 4.3: Belief State Tracking

From Problems

Setup

- Four rooms: A, B, C, D. Package is in exactly one.
- Prior: $b_0(X) = P(\text{Package in } X) = 0.25$ for each X.
- Scanner: $P(\text{beep} \mid \text{package here}) = 0.9$, $P(\text{beep} \mid \text{no package}) = 0.05$.

Part 1: Room A, No Beep

The robot enters Room A and the scanner does **not** beep. Let $E_1 = \overline{\text{beep}}$ denote this evidence. We update the belief for each hypothesis about where the package is.

Likelihoods of "no beep in Room A" under each hypothesis:

- Package in A: $P(E_1 \mid A) = 1 - 0.9 = 0.10$ (scanner missed it)
- Package in B: $P(E_1 \mid B) = 1 - 0.05 = 0.95$ (no package here, no beep is expected)
- Package in C: $P(E_1 \mid C) = 0.95$ (same reasoning)
- Package in D: $P(E_1 \mid D) = 0.95$ (same reasoning)

Unnormalized posteriors $P(E_1 \mid X)\, b_0(X)$:

Room X	$P(E_1 \mid X)$	$b_0(X)$	Unnormalized
A	0.10	0.25	0.025
B	0.95	0.25	0.2375

Room			Unnormalized
C	0.95	0.25	0.2375
D	0.95	0.25	0.2375

Normalizing constant: $Z = 0.025 + 3 \times 0.2375 = 0.025 + 0.7125 = 0.7375$

Normalized posteriors $b_1(X)$:

$$b_1(A) = \frac{0.025}{0.7375} \approx 0.0339$$

$$b_1(B) = b_1(C) = b_1(D) = \frac{0.2375}{0.7375} \approx 0.3220$$

Check: $0.0339 + 3 \times 0.3220 = 0.0339 + 0.9661 = 1.0000 \checkmark$

The no-beep in Room A dramatically reduced $b(A)$ from 25% to 3.4%, and the probability mass shifted roughly equally to the other three rooms.

Part 2: Room B, Beep

Starting from b_1, the robot enters Room B and the scanner **beeps**. Let $E_2 = \text{beep}$.

Likelihoods of "beep in Room B" under each hypothesis:

- Package in A: $P(E_2 \mid A) = 0.05$ (no package in B, false positive)
- Package in B: $P(E_2 \mid B) = 0.90$ (package is here, true positive)
- Package in C: $P(E_2 \mid C) = 0.05$ (no package in B, false positive)
- Package in D: $P(E_2 \mid D) = 0.05$ (no package in B, false positive)

Unnormalized posteriors $P(E_2 \mid X)\, b_1(X)$:

Room X	$P(E_2 \mid X)$	$b_1(X)$	Unnormalized
A	0.05	0.0339	0.001695
B	0.90	0.3220	0.28983
C	0.05	0.3220	0.01610
D	0.05	0.3220	0.01610

Normalizing constant: $Z = 0.001695 + 0.28983 + 2 \times 0.01610 = 0.32373$

Normalized posteriors $b_2(X)$:

$$b_2(A) = \frac{0.001695}{0.32373} \approx 0.0052$$

$$b_2(B) = \frac{0.28983}{0.32373} \approx 0.8953$$

$$b_2(C) = b_2(D) = \frac{0.01610}{0.32373} \approx 0.0497$$

Check: $0.0052 + 0.8953 + 2 \times 0.0497 = 0.9999 \approx 1.0 \checkmark$

After two observations, the robot is about 90% confident the package is in Room B. Room A has been nearly eliminated (0.5%), and Rooms C and D each have about 5%.

Part 3: Rooms to 99% Confidence (Worst Case)

The question is: how many rooms must the robot visit before it can identify the package location with at least 99% confidence, in the worst case?

Key observations: With noisy sensors, no single observation can give certainty. Each no-beep reduces a room's posterior but doesn't eliminate it. Each beep boosts a room's posterior but doesn't confirm it. The worst case for the number of visits occurs when the package is in the *last* room the robot checks, so the robot accumulates only no-beep evidence for the first several visits.

Scenario: The robot visits A, B, C, D in order. The package is in D. Assume the robot gets the most likely observation at each step (no-beep when the package is absent, beep when present).

After visiting A (no beep): From Part 1, $b(A) \approx 0.034$, $b(B) = b(C) = b(D) \approx 0.322$.

After visiting B (no beep): The likelihood of no-beep-in-B is 0.10 if the package is in B (false negative) and 0.95 if the package is elsewhere. So B's posterior drops sharply, while all other rooms' posteriors increase slightly after normalization:

- Unnormalized: A: $0.95 \times 0.034 = 0.032$, B: $0.10 \times 0.322 = 0.032$, C: $0.95 \times 0.322 = 0.306$, D: $0.95 \times 0.322 = 0.306$
- Normalizing ($Z = 0.676$): $b(A) \approx 0.048$, $b(B) \approx 0.048$, $b(C) \approx 0.452$, $b(D) \approx 0.452$.

Note that $b(A)$ went *up* (from 0.034 to 0.048). This is correct: a no-beep in B is expected if the package is in A, so the observation is mildly favorable for A. After normalization, A gains a small share from B's loss.

After visiting C (no beep):

- Unnormalized: A: $0.95 \times 0.048 = 0.046$, B: $0.95 \times 0.048 = 0.046$, C: $0.10 \times 0.452 = 0.045$, D: $0.95 \times 0.452 = 0.429$
- Normalizing ($Z = 0.566$): $b(A) \approx 0.080$, $b(B) \approx 0.080$, $b(C) \approx 0.080$, $b(D) \approx 0.760$.

After 3 visits (all no-beep), the robot is 76% confident the package is in D—the only unvisited room. Not yet 99%.

After visiting D (beep):

- Unnormalized: A: $0.05 \times 0.080 = 0.0040$, B: $0.05 \times 0.080 = 0.0040$, C: $0.05 \times 0.080 = 0.0040$, D: $0.90 \times 0.760 = 0.6840$
- Normalizing ($Z = 0.696$): $b(D) \approx 0.983$. Close but not yet 99%.

After visiting D again (second beep):

- Unnormalized: D: $0.90 \times 0.983 = 0.885$, others: $0.05 \times (\sim 0.006) \approx 0.0003$ each.
- Normalizing: $b(D) \approx 0.999$. Now past 99%.

Answer: **5 visits** (A, B, C, D, D again). Visiting all 4 rooms once yields $\approx 98\%$ confidence; one additional visit to the room that beeped pushes past 99%. With noisy sensors, repeated evidence is needed to reach high confidence—a single beep is not enough because it could be a false positive.

Part 4: Perfect Scanner

With $P(\text{beep} \mid \text{package here}) = 1$ and $P(\text{beep} \mid \text{no package}) = 0$:

No beep in Room A: $P(\text{no beep} \mid \text{package in A}) = 0$, so the likelihood is zero. After normalization:

$$b_1(A) = 0, \quad b_1(B) = b_1(C) = b_1(D) = \frac{1}{3}$$

The package is *definitely* not in A. One observation eliminates a room entirely.

Beep in Room B: $P(\text{beep} \mid \text{package in B}) = 1$ and $P(\text{beep} \mid \text{package elsewhere}) = 0$. So:

$$b_2(B) = 1, \quad b_2(\text{others}) = 0$$

One beep identifies the room with certainty.

Connection to the logical agent (Chapter 3): With a perfect scanner, the Bayesian update reduces to logical elimination. A no-beep *entails* that the package is not in the current room (the hypothesis is ruled out with probability 0, not merely reduced). A beep *entails* that the package is here (probability 1). This is exactly how the knowledge-based agent from Chapter 3 operates: each percept narrows the set of possible worlds until only one remains. The probabilistic framework subsumes the logical one as a special case—when sensors are perfect, posteriors collapse to 0 or 1, and Bayesian reasoning becomes logical deduction.

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